

SECTION A**1 (short)**

- (a) Calculate and sketch the Thévenin and Norton equivalents of the circuit of Fig. 1. [6]
- (b) A load resistor R_L is connected across terminals A and B of the circuit of Fig. 1. Calculate the value of R_L that achieves maximum power transfer and calculate the power dissipated in the load. [4]

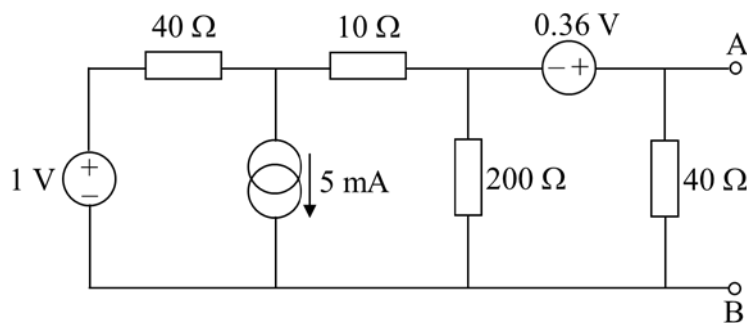


Fig. 1

2 (short) Figure 2 shows a simple induction heater used in a domestic kitchen stove top. The heater coil is modelled as an inductor of $90\ \mu\text{H}$ and it is wired in parallel with a capacitor C .

- (a) Calculate the capacitance, C , required to achieve a resonant frequency of 24 kHz. [4]
- (b) The capacitance C is set to 100 nF and the power supply is set to apply v_{in} with r.m.s. voltage of 200 V at a frequency of 24 kHz. Find the peak amplitude of the current through the resistor and its phase with respect to the voltage v_{in} . [6]

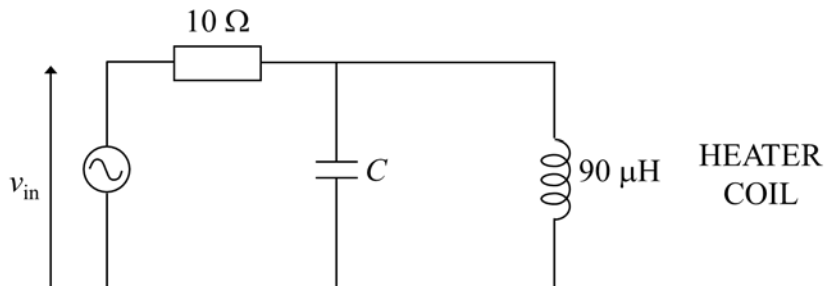


Fig. 2